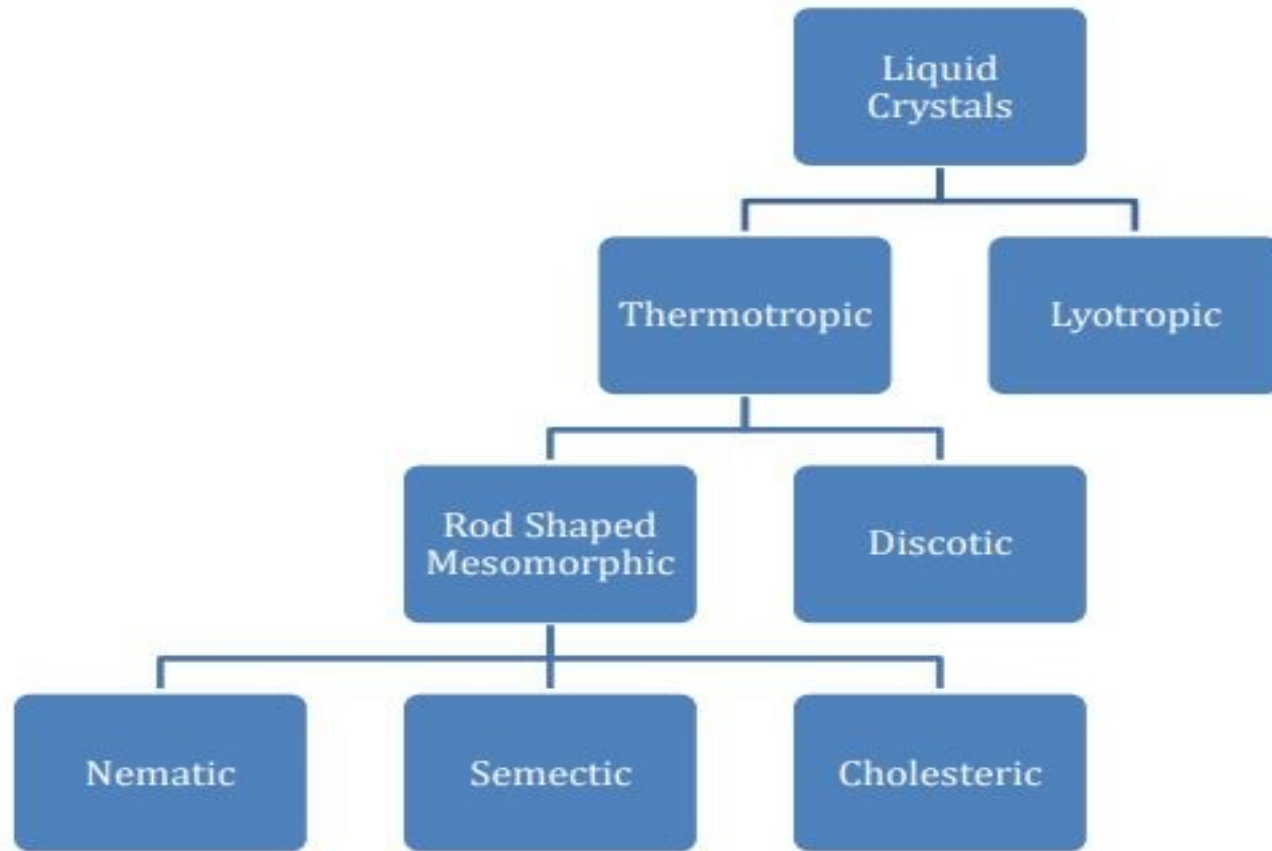


ENGINEERING MATERIALS

Liquid Crystals

4 times for 2-5 marks

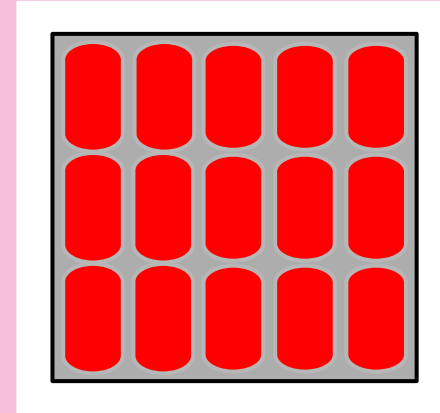
- A crystalline solid does not melt directly into liquid state but passes first through an intermediate change that is called *liquid crystal*.
- Small electric field can disturb the alignment of molecules.
- Two types: Thermotropic and Lyotropic.
- Solid materials heated to produce LCs are thermotropic.
- Dissolving solutes in liquid solutions to increase its concentration to reach LC stage are lyotropic.



4 times for 2-5 marks

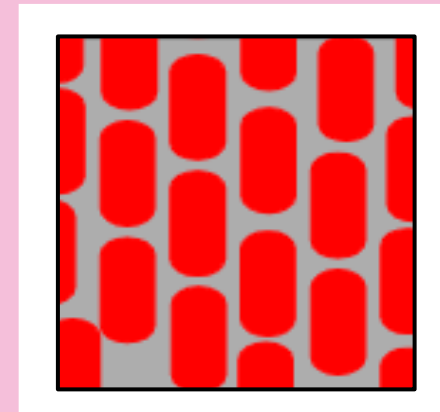
A. Smectic

- Orientation and periodicity is retained.
- Have ordered position and orientation.



B. Nematic

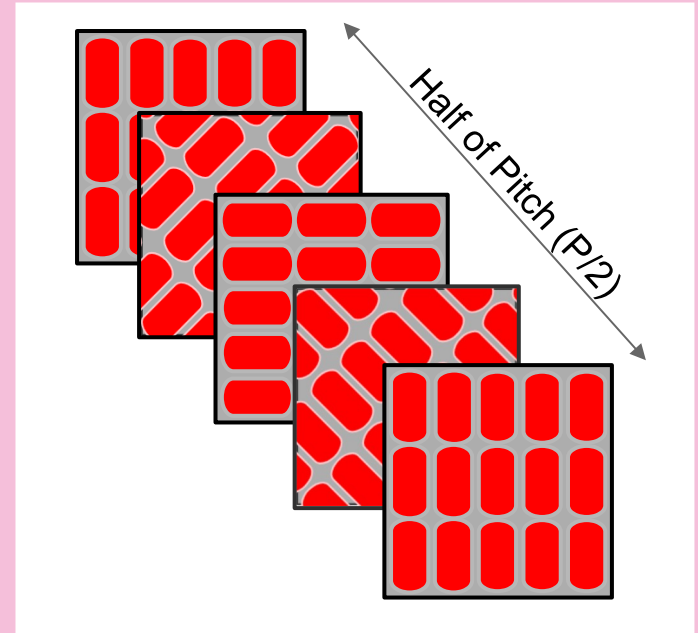
- Orientation is retained
- Periodicity is lost
- Have ordered orientation
- Disordered position



C. Cholesteric

- Same arrangement of molecules as that of Nematic type
- Consists of *parallel* layers.
- *Helical* structure
- Recurring design in animal and plant kingdoms.
- **Functions:** maximisation of packing efficiency, optical information, radiation protection, mechanical stability.

4 times for 2-5 marks



Applications Of Liquid Crystals.

4 times for 2-5 marks

- Used in *Liquid Crystal Displays* (LCD):
 - digital watches
 - pocket calculators

Due to *ease of change in orientation* made by electric field or pressure.

- Used in *gas-liquid chromatography* for their *electrical properties* which lie between crystalline solids and isotropic liquid.
- Used in high speed computing.

Properties Of Ferroics

Ferroics is the name given to the study of Ferromagnetism, Ferroelectric and Ferroelasticity.

1. **Ferromagnetism:** A magnetism that is controlled by application of magnetic field.
2. **Ferroelectricity:** An electric polarization that is switchable by applied electric field.
3. **Ferroelasticity:** A deformation that is switchable by applied stress.

Multiferroics

1 time for 5 marks

Multiferroics are materials that exhibit more than one of the above primary properties of the Ferroics materials. They are of two types:

TYPE I

- Ferroelectricity and ferromagnetism occur at different temperatures and with different mechanisms.
- Ferroelectricity and ferromagnetism in such materials is largely independent of one another.
- Eg: YMnO_3

TYPE II

- Ferroelectricity and ferromagnetism occur at same temperatures and with same mechanisms.
- These materials' magnetism causes ferroelectricity implying strong coupling between the two.
- Eg: TbMnO_3

Applications Of Multiferroics

1 time for 5 marks

- Electric field control of magnetism
- Spintronic devices
- Microelectronic devices MOSFET
- Information storage devices
- Quantum electromagnets in Particle physics
- Sensors
- Radio and high frequency devices

Magneto Resistive Materials

1 time for 3 marks

The materials that have a tendency to change the value of their electrical resistance in an externally applied electric field.

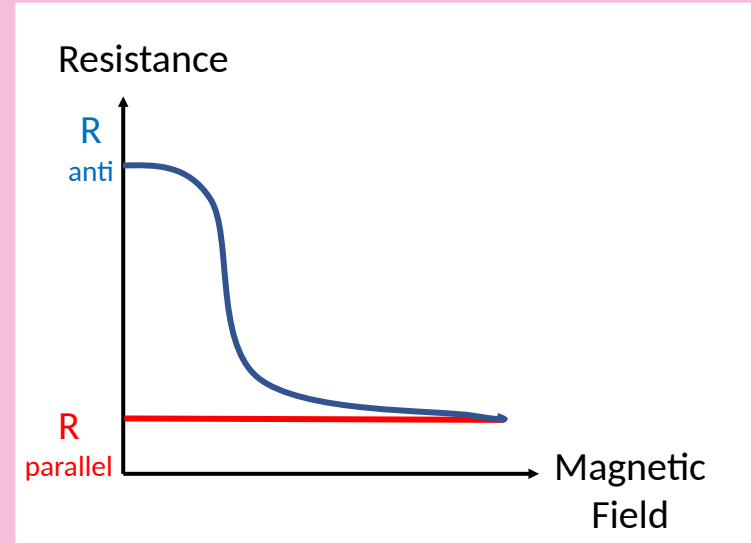
The tendency of such materials is known as magnetoresistance.

Two main types of Magnetoresistance:

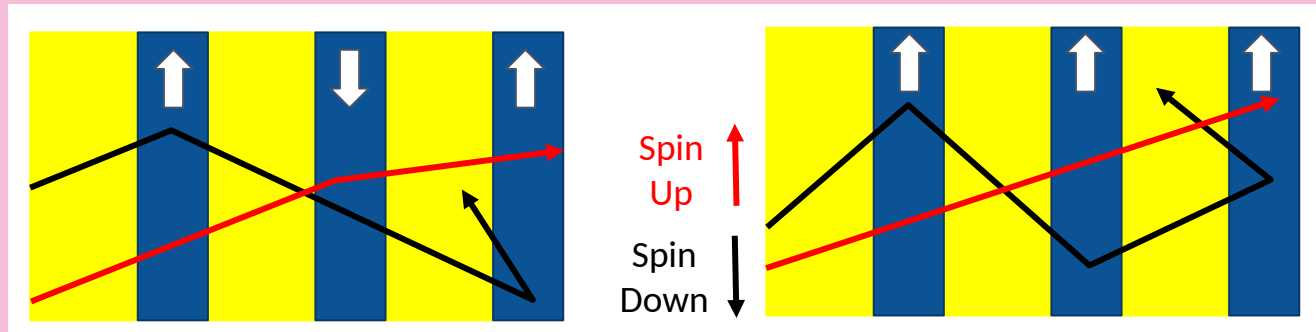
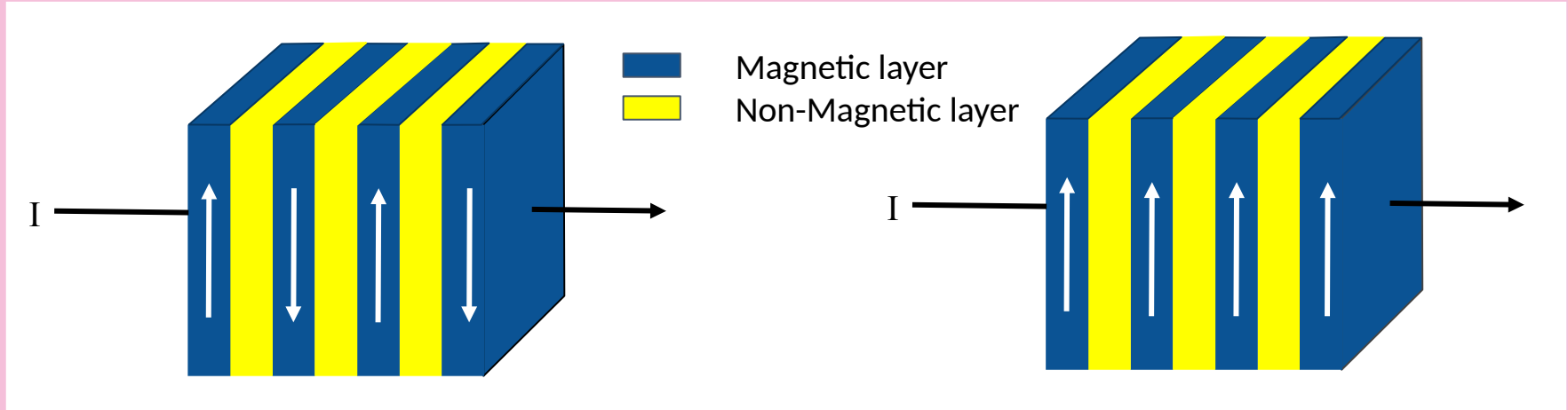
- GMR: Giant magneto resistance
- CMR: Collosal magneto resistance

Giant Magnetoresistance (GMR)

- Magnetoresistance effect observed in layers of alternating ferromagnetic and non-magnetic layers.
- Scattering of electrons with spin value (small magnets) in a ferromagnetic material reduces current which increases electrical resistance.
- Magnetization of system becomes saturated which results in drop of electrical resistance.
- Electric resistance is *low* for *parallel alignment* and *high* for *antiparallel alignment* of ferromagnetic layers.
- **Applications:** Sensors to read data, Hard disc, Drives, Biosensors, MEMS

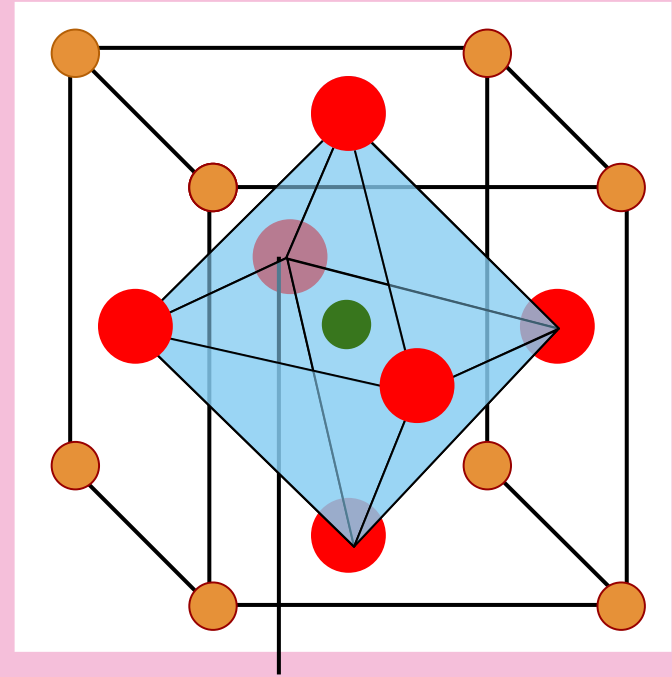


Giant Magnetoresistance (GMR)



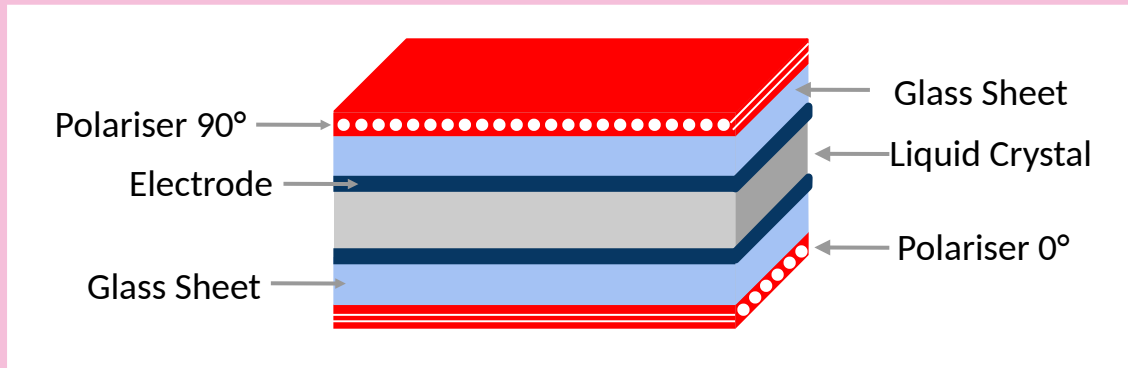
Colossal Magnetoresistance (CMR)

- The tendency of a material to demonstrate drastic change in their electrical resistance, in the presence of a external magnetic field.
- A property of a few manganese based perovskite oxides that drastically changes their electrical resistance in presence of a strong magnetic field.
- **Applications:**
 - Read devices
 - Bolometers
 - Spintronics
 - Magnetic field sensors.



Liquid Crystal Display (LCD)

- Made of: Thin layer of liquid crystal, Two glass sheets, Transparent electrodes.
- **WORKING:**
 - Illumination required from external source
 - Use of cholesteric property
 - Indium Tin oxide layer coated on glass sheet acts as electrode
 - Voltage applied at electrode aligns mesogens
 - Light waves get aligned along the field
 - Hence no transmission of light possible



Liquid Crystal Display (LCD)

- **Merit:** Requires Power in Watts
- **Demerit:** Activated by AC and Display is destroyed by DC
- **Application:**
 - Calculators
 - Computers
 - TVs

